and 3 times from those that share F_{1b}. This process guarantees variety of the concepts and targets.

4 Agent Roles

Below we describe how we employ ChatGPT as game players to generate the dialogues and as diagnostic agents to evaluate the Questioner's information seeking ability.

4.1 Game Players

To generate the dialogues, ChatGPT is instructed to play the role of the Questioner (**ChatGPT-Q**) and of the Answerer (**ChatGPT-A**) with a similar system prompts. The shared part of the prompts explicitly states the only possible answers are 'yes'and 'no'. ChatGPT-Q is told to ask as few questions as possible; the Questioner starts by asking the first question, which is appended to the Answerer's prompt in order to generate the first answer. In this way, the dialogue history is iteratively increased after each turn. ChatGPT-A is told to acknowledge when the Questioner has correctly guessed the item by answering "Yes! That's correct.". Focusing on ChatGPT's capabilities of reasoning about the hypothesis space and asking questions that reflect such reasoning, we retain only successful dialogues. More precisely, the dialogue is kept if the Answerer considers the target reached. Our evaluation is focused on the Questioner role, hence, for it we define theoretically an upper and a lower bound as described below.

We take as upper-bound a model that similarly to adults seeks for a property shared by several items in the space. In particular, we use the **optimal agent** which acts similarly to a binary search algorithm: at each turn, it divides in half the hypothesis space under discussion ($N/2$). When only two items are left, the optimal agent makes a guess that has the 50% chance of being the correct target. This half-split strategy takes on average $\log_2 N + 1/2$ turns to solve the game, where N is the number of items at the beginning of the game.

As lower-bound we consider a model close to the 4-Y child who tends to scan the space item by item. Therefore, our **baseline agent** acts similarly to a linear search algorithm: at each turn, it divides the space into $1\ vs.\ N-1$. Given N items at the beginning of the game, it takes on average $N/2$ turns to solve the game.

4.2 Diagnostic Agents

To evaluate the model's ability to stepwise reduce the hypothesis space we exploit ChatGPT in the role of an external Oracle (ChatGPT-Oracle), and of an external Guesser (ChatGPT-Guesser). Moreover, we activate the Guesser internal to the Questioner by prompting the model to update the candidates at each turn (ChatGPT-Q-stepwise).

ChatGPT-Oracle is given a question in the dialogue sets described in the previous section and for each item in the hypothesis space of the corresponding game says whether the item has or does not have the required property.[5] This provides us with Y/N-annotation of the hypothesis space that we use to obtain a "ground truth" updated space at each turn. The feasibility of such method relies on the fact that the dialogues are rather simple and no actual linguistic dependencies are in place between the turns (See Supplementary Material for details). **ChatGPT-Guesser** is given chucks of the dialogue histories generated by the game players and is asked to list the candidates till the given turn. Finally, we modify the prompt of ChatGPT-Q by asking it to list the candidates under discussion stepwise before asking a new question (**ChatGPT-Q-stepwise**). The prompts used for each role are reported in the Supplementary Material.

5 Experimental Setup

We are interested in understanding whether ChatGPT's language generation is driven by its reasoning process. To answer this question, we propose a number of measures aimed to shed light on the reasoning processes that are implicit in the game: identify the high-level property shared by several items, update the space stepwise, and efficiently arrive to a space with just one possible candidate and realize that it is time to stop asking questions. Not having the possibility to run an ablation study of the model, we simulate it by comparing ChatGPT-Q, simply prompted to play the game, with ChatGPT-Q-stepwise which is explicitly told to update the space turn by turn.

Information seeking strategy Following the method used in the Cognitive Science literature to evaluate children's developmental phases, we evaluated the information seeking strategy used

[5] We verified the reliability of the annotation by evaluating the model's accuracy on a sample of 180 questions: it correctly answered 83% of the questions.

by ChatGPT-Q by observing the type of questions it asks and their informativeness. First of all, we compute the percentage of questions that are *Hypothesis-Scanning (HS)* and *Constraint-Seeking (CS)*. A question is considered *HS* iff it explicitly mentions one of the candidates in the hypothesis space. All the other questions are considered *CS*. We compute the percentage of HS and CS questions within a game and by the position of the turn within the dialogue.[6]

Following Ruggeri et al. (2016), Meder et al. (2019) and Testoni (2023), we compute the *Expected Information Gain (EIG)* of each question and report the average EIG per turn.[7] As clearly explained in Meder et al. (2019), the information gain of a question is the entropy in the space (given by the number of items and the associated probability) at turn t_i before asking the question minus the expected entropy after asking it (t_{i+1}):

$$IG = H_{t_i} - H_{t_{i+1}}$$

As in Meder et al. (2019), in our case, we consider all items in the space to be equally likely to be the target. Hence, what defines entropy is the number of items within the subsets answered with Yes vs. No, based on the external Oracle annotation.

A model that asks fewer HS, especially in the earlier turns is closer to the more efficient strategy used by adults. Its question EIG is expected to be very high in the early turns and to decrease in the later ones.

Hypothesis space update A core skill of the Questioner playing the 20-Q game is the ability to *mentally* keep the space of the hypothesis updated stepwise. We evaluate whether ChatGPT-Q is able to update at each turn the hypothesis space based on the given dialogue history. Again, we consider the Yes/No-annotation obtained from ChatGPT-Oracle as the ground truth and compute the hypothesis space at turn $HypSp_{t_i}$ by filtering out from $HypSp_{t_{i-1}}$ the items which do not have the property required at t_i. We compare the ground truth Hypothesis Space with a) the one generated by the external Guesser, ChatGPT-Guesser, and b) the one generated by the Questioner itself when prompted to explicitly update the list of candidates stepwise (ChatGPT-Q-stepwise). To this end, we compute the symmetric difference between the generated sets with the ground-truth ones for every question, and report the average symmetric difference of each game turn. The symmetric difference between two sets A and B is denoted by $A\Delta B$ and is defined as follows:

$$A\Delta B = (A - B) \cup (B - A)$$

For ChatGPT-Guesser, a high difference would mean that the model has difficulty integrating the information collected through the dialogue history. While for ChatGPT-Q-stepwise it would signal a difficulty in integrating the answer with the question turn by turn.

Search efficiency We measure the efficiency of the ChatGPT-Q's game strategy by computing the *average number of questions per game, (AQ)*. In addition to this, and as in (Ruggeri et al., 2016), we consider a question *unnecessary (UQ)*, if the preceding dialogue history already contained the information to identify the target. Again, we use the Y/N-annotation by ChatGPT-Oracle to determine whether this point has been reached by ChatGPT-Q.

The more the search strategy is effective, the shorter is the dialogue. The higher the number of UQ the closer is the model to pre-scholar children, who have been shown not to have learned the stopping rule yet. If the model asks just one UQ as last turn, that would still qualify it adult-like, since adults have been shown to ask a confirmation question before making the final guess (Testoni et al., 2022).

Experimental Settings We expect that bigger candidate sets could challenge the model's capacity to keep track of the information obtained through the dialogue, since they might require longer interactions. Moreover, with the GPT-based 8 cds the model should have all the knowledge to quickly arrive to identify the target. Hence, if ChatGPT's knowledge properly drives its question generation the dialogues of the games based on it should display an almost optimal information seeking strategy. Based on these conjectures, we compare the model when playing games whose hypothesis space a) consists of 8 and 16 candidate sets, b) is built out of McRae- or ChatGPT-feature norms. If the question generation is driven by the reasoning process on the space, we expect the model's

[6] A third type of questions are the *Pseudoconstraint-seeking (PCS)* which ask about a property but actually refer to only one item among the candidates. For the sake of simplicity, we do not consider them in our analysis, but see the Supplementary Material for statistics on them.

[7] We computed the EIG adapting the code by Testoni (2023).

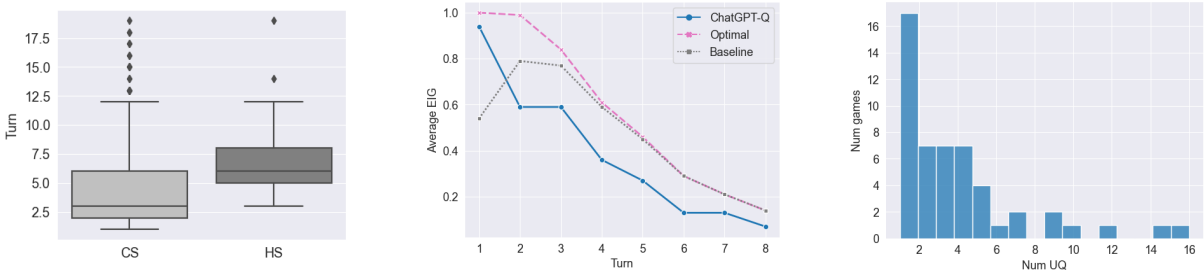

Figure 2: **Left**: ChatGPT-Q asks CS questions mostly in the earlier turns and HS in the later ones. Mann-Whitney U test shows the difference is statistically significant $p < 0.001$; **Middle**: ChatGPT's EIG is almost as high as the optimal model's EIG at the first turn, but it is lower than of the baseline's EIG at later turns. **Right**: Distribution of unnecessary questions.

performance to decrease when challenged with a higher number of candidates and to increase when the hierarchical structure of the candidates is based on the model's internal knowledge.

6 Results

In this section, we show the results we obtained following the experimental setup defined in Section 5. We start by evaluating ChatGPT's performance on the games with 8 candidates selected with the McRae-norms (8-McRae), and we then move to compare these results with those obtained by the model when challenged with spaces containing an higher number of candidates (16-McRae), or whose nested structure reflects the model's knowledge representation (8-GPT). Finally, we move to evaluate ChatGPT on the 8-McRae games when asked to play the game (ChatGPT-Q) and when asked to explicitly update the hypothesis space stepwise while playing the game (ChatGPT-Q-stepwise).

6.1 ChatGPT-Q on the McRae-8 games

Through the measures introduced above, here we aim to take a picture of how well and efficiently ChatGPT searches for information by considering McRae-8 games.

Information seeking strategy The results for the type of questions asked by ChatGPT-Q, the *optimal agent* and the *baseline* can be seen in Table 1. By construction, the optimal agent asks $log_2 N - 1$ CS question per game, followed by 1 or 2 HS questions (hence 1.5 on average), until it guesses the target; in other words, 57.14% of its questions are CS, and 42.86% are HS. Instead, the baseline asks only HS questions and, on average, it guesses the target in 4 questions. ChatGPT-Q asks mostly Constraint-Seeking questions (73.77%), it tends to ask CS questions in the early turns and HS questions towards the end of the dialogue – when indeed the latter becomes more efficient to split the space (Figure 2, left).

8 cds based on McRae-norms			
	HS	CS	AQ
Optimal	42.86	57.14	3.50
Baseline	100	0	4.00
ChatGPT	26.33	73.77	7.24

Table 1: Information seeking strategy: Upper and lower bound of the overall percentage of hypothesis scanning (HS) vs. constraint seeking (CS) questions, and the average number of questions per game (AQ) -the difference is statistically significant based on the Wilcoxon signed-rank test (e.g., wrt the optimal agent, $p < 0.001$.

Moreover, our results show that the EIG of ChatGPT-Q's questions through the dialogue is far from the optimal agent's EIG (that half-split the space at each turn) and even lower than the baseline's (that splits the space into 1 vs. the other candidate at each turn) (see Figure 2, middle). Summing up, on the surface level, the strategy used by ChatGPT-Q reflects what an adult would do. However, the EIG analysis shows that ChatGPT-Q asks more uninformative questions compared both to the optimal agent and the baseline.

Hypothesis Space update We evaluate whether ChatGPT-Guesser is able to list the candidates that are still possible candidates based on the Question-Answer exchanges between ChatGPT-Q and ChatGPT-A. We do so by computing the difference, turn by turn, of such list with those considered as "ground truth" based on ChatGPT-Oracle annotations. The pattern we find (see the blue line in Figure 3, right) suggests the model has difficulty in integrating the information collected through the